

# **JAVA VITYARTHI PROJECT**

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**PROJECT TITLE:**

**DIGITAL FOOTPRINT ANALYZER &  
PRIVACY RISK ASSESSMENT SYSTEM**

## 1. Introduction

Today, we use the internet and social media for almost everything. We create accounts, share photos, communicate with others, use different apps, and often share information without thinking much about it. All this information becomes part of our **digital footprint**.

A digital footprint can include things like usernames, photos, email addresses, phone numbers, location, and other personal details. If too much of this information is publicly available, it can create privacy and security risks.

The **Digital Footprint Analyzer & Privacy Risk Assessment System** was developed to help users understand these risks. The application asks simple questions about their online habits and uses the answers to calculate a privacy risk score. It also provides recommendations to help users improve their privacy.

## 2. Problem Statement

Many users are not aware of how much personal information they are sharing online or which of their habits may create privacy risks.

The purpose of this project is to develop a simple system that can:

- Analyze common online privacy habits.
- Calculate an overall privacy risk score.
- Identify the main areas where risk exists.
- Provide useful recommendations.
- Help users track their privacy results over time.

## 3. Objectives

The main objectives of this project are:

- To develop an easy-to-use privacy assessment system.

- To identify common digital privacy risks.
- To calculate risk and privacy scores.
- To provide personalized recommendations.
- To show how certain privacy improvements can affect the risk score.
- To maintain a local history of previous assessments.
- To create awareness about digital privacy.

## 4. Project Scope

The current project works as a privacy self-assessment tool.

The user answers questions about their online habits, and the system analyzes the answers. It does not automatically scan social media accounts or search the internet for personal information.

The application also does not ask users to enter passwords, OTPs, or account login details. This keeps the assessment simple and avoids collecting sensitive credentials.

## 5. Technologies Used

| Technology    | Purpose                          |
|---------------|----------------------------------|
| Java          | Main programming language        |
| Java Swing    | Creating the graphical interface |
| CardLayout    | Moving between different screens |
| File Handling | Saving and reading data          |
| CSV           | Storing assessment history       |

The project uses standard Java and does not require external libraries.

## 6. System Workflow

The basic working of the application is:

**User Input → Privacy Assessment → Risk Calculation → Risk Level → Recommendations → Report**

First, the user completes the assessment. The system then processes the answers, calculates the risk for different categories, and generates the final result.

## 7. Privacy Assessment

The assessment contains questions related to the user's online activities and privacy habits.

The system considers:

- Number of public social profiles
- Use of the same username on different platforms
- Unknown followers
- Identifiable photos
- Public email address
- Public phone number
- Public birthday
- Public education details
- Location sharing
- Live location sharing
- Public travel plans
- Password reuse
- Two-factor authentication
- Privacy settings review
- Unused accounts
- App permissions

The user provides only the information needed for the assessment and does not enter actual account credentials.

## **8. Risk Categories**

The system divides the assessment into six categories.

### **8.1 Social Media Exposure**

This category looks at public social media activity, including public profiles, usernames, followers, and identifiable photos.

### **8.2 Account Security**

This category focuses on password reuse and the use of two-factor authentication.

### **8.3 Location Privacy**

This category considers location sharing, live location sharing, and public travel plans.

### **8.4 Personal Information**

This category covers personal details such as email, phone number, birthday, and education information.

### **8.5 Account Hygiene**

This category looks at unused or old online accounts.

### **8.6 App Privacy**

This category considers the permissions given to applications.

## 9. Risk Calculation

Each risk category has a different weight in the final calculation.

| Category              | Weight |
|-----------------------|--------|
| Social Media Exposure | 20%    |
| Account Security      | 25%    |
| Location Privacy      | 20%    |
| Personal Information  | 15%    |
| Account Hygiene       | 10%    |
| App Privacy           | 10%    |

The category scores are combined according to these weights to calculate the overall risk score.

The privacy score is calculated using:

$$\text{Privacy Score} = 100 - \text{Risk Score}$$

For example, if the risk score is 60, the privacy score will be 40.

## 10. Risk Levels

The final risk score is converted into a simple risk level.

### Score Risk Level

0–20 LOW

21–40 MODERATE

41–60 MEDIUM

61–80 HIGH

81–100 CRITICAL

This makes the result easier to understand instead of showing only a number.

## **11. Recommendation System**

After calculating the result, the application gives recommendations based on the user's answers.

For example, the system may suggest:

- Enable two-factor authentication.
- Avoid using the same password on multiple accounts.
- Review social media privacy settings.
- Reduce unnecessary location sharing.
- Limit personal information visible to the public.
- Review app permissions.
- Check unused accounts.

The recommendations are generated using predefined rules in the application.

## **12. Dashboard and Report**

The dashboard provides a quick summary of the assessment.

It shows:

- Risk score
- Privacy score
- Risk level
- Category-wise results
- Recommendations

The application also provides a detailed report that gives more information about the assessment. The report can be exported as a text file for future reference.

Dashboard screenshots:

# Digital Footprint Analyzer

## Privacy Risk Assessment System

Understand how much personal information you expose online.

The system evaluates your digital footprint across:

Social Media • Account Security • Location Privacy  
Personal Information • Account Hygiene • App Privacy

No passwords, OTPs or private account credentials are required.

[Start Privacy Assessment](#)

Activate

## Privacy Risk Dashboard

Overview of your current digital footprint.

### OVERALL RISK

# 13

out of 100

**Privacy Score: 87/100**

**Risk Level: LOW**

### CATEGORY-WISE RISK

Social Media Exposure

0/100

Account Security

40/100

Location Privacy

0/100

Personal Information

0/100

Account Hygiene

5/100

App Privacy

25/100

[New Assessment](#)[Risk Simulator](#)[Detailed Report](#)

Activate

Digital Footprint Analyzer

## Privacy Assessment

Answer the following questions honestly.

1. SOCIAL MEDIA EXPOSURE

Number of public social media profiles: 

2

☐

I use the same username on multiple platforms.

☐

I accept followers/friend requests from unknown people.

☐

My photos may reveal identifiable personal information.

2. PERSONAL INFORMATION EXPOSURE

☐

My email address is publicly visible.

☐

My phone number is publicly visible.

☐

My complete birthday/date of birth is publicly visible.

☐

My school/college information is publicly visible.

3. LOCATION PRIVACY

☐

My location information is publicly visible.

☐

I sometimes share my live/current location.

☐

I publicly share travel plans before/during travel.

4. ACCOUNT SECURITY

☐

I reuse passwords across multiple accounts.

☐

I use two-factor authentication (2FA).

Reset

Analyze My Digital Footprint

Activate

Digital Footprint Analyzer

## Detailed Privacy Report

Identified risks and personalized recommendations.

OVERALL RESULT

Risk Score : 13/100

Privacy Score : 87/100

Risk Level : LOW

CATEGORY ANALYSIS

Social Media Exposure : 0/100

Account Security : 40/100

Location Privacy : 0/100

Personal Information : 0/100

Account Hygiene : 5/100

Application Privacy : 25/100

IDENTIFIED RISKS

1. Two-factor authentication is not enabled.

2. Privacy/security settings are not regularly reviewed.

3. Application permissions have not been reviewed.

PERSONALIZED RECOMMENDATIONS

1. Enable Two-Factor Authentication

Back to Dashboard

Export Report

Activate

### 13. Assessment History

The application saves previous assessment summaries in a local CSV file.

The stored information includes:

- Date
- Risk score
- Privacy score
- Risk level

This allows users to compare their results over time and see whether their privacy habits are improving.

### 14. Risk Simulator

The **Risk Simulator** allows users to test possible privacy improvements without changing their original assessment.

For example, users can simulate:

- Enabling two-factor authentication
- Stopping location sharing
- Hiding a public phone number
- Stopping password reuse

The system then calculates the simulated risk score.

This feature helps users understand how different privacy actions can affect their overall result.

### 15. Privacy Education

The application also contains a small educational section about digital privacy.

It covers topics such as:

- Digital footprints
- Password security
- Two-factor authentication
- Social media privacy
- Location privacy
- App permissions

This section helps users understand the reason behind some of the recommendations given by the application.

## 16. Testing

The main features of the application were tested using different inputs.

| Test Case                | Expected Result                  |
|--------------------------|----------------------------------|
| Launch application       | Welcome screen appears           |
| Start assessment         | Assessment screen opens          |
| Complete assessment      | Risk score is calculated         |
| View dashboard           | Assessment results are displayed |
| View report              | Detailed result is shown         |
| Generate recommendations | Relevant suggestions appear      |
| Save assessment          | History is updated               |
| Open simulator           | Simulation options are displayed |
| Export report            | Text report is created           |

The testing was used to check the main screens, calculations, recommendations, history, simulator, and report export functionality.

## 17. Limitations

The current version of the project has some limitations:

- It does not automatically scan online platforms.
- It does not connect to social media APIs.
- It uses predefined rules instead of machine learning.
- The result depends on the information provided by the user.
- Assessment history is stored locally using CSV.

These limitations can be addressed in future versions.

## **18. Future Scope**

The project can be further improved by adding:

- Machine learning-based risk analysis.
- Database support.
- Web and mobile versions.
- Integration with legitimate APIs for automated analysis.
- Graphs and more detailed analytics.
- Privacy reminders.
- More advanced risk calculations.

## **19. Project Structure**

Digital-Footprint-Analyzer/

DigitalFootprintAnalyzer.java  
 digital\_footprint\_history.csv  
 README.md  
 Screenshot

The main application is contained in

- DigitalFootprintAnalyzer.java.

## **20. How to Run the Project**

A compatible Java Development Kit (JDK) is required.

Open the project folder and compile the program using:

```
javac DigitalFootprintAnalyzer.java
```

Then run it using:

```
java DigitalFootprintAnalyzer
```

The application will open with the welcome screen.

## **21. Conclusion**

The Digital Footprint Analyzer & Privacy Risk Assessment System is a simple application designed to help users understand their online privacy habits.

The system collects information through a questionnaire, analyzes the answers using predefined rules, and provides a risk score, privacy score, risk level, and recommendations.

Features such as the dashboard, assessment history, detailed report, privacy education, and risk simulator make the application more useful and interactive.

The project also follows a privacy-friendly approach by not asking users for passwords, OTPs, or account credentials.

Overall, this project combines Java programming, GUI development, file handling, rule-based decision making, and digital privacy awareness into one practical application.

## **22. Declaration**

I hereby declare that the project titled “Digital Footprint Analyzer & Privacy Risk Assessment System” has been developed as part of my academic project work.

The project has been developed using Java and Java Swing and focuses on assessing common digital privacy habits using a rule-based risk assessment system.

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